

Pairing Proofs and Polynomial Ring Toolkit

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Abstract. Elliptic curve pairings are fundamental to modern zero-knowledge proof systems, including Groth16, PLONK, and KZG polynomial commitments. In resource-constrained execution environments (Ethereum VM, BitVM, blockchain smart contracts) where computation is metered by gas fees, and in arithmetized computation systems (R1CS, Plonkish, Cairo AIR) where each computational step imposes significant prover overhead, the cost of pairing computation remains a practical bottleneck.

We present a public-coin interactive proof system that consolidates the Miller loop iterations and final exponentiation into a single polynomial relation. The relation can then be verified as polynomial identities greatly reducing the pairing costs.

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Keywords: Elliptic curve pairings · Zero-knowledge proofs · Polynomial identity testing · Blockchain verification · Recursive proofs.

1 Introduction

Blockchain technology continues to mature with improving scalability and infrastructure. However, privacy remains a significant barrier to entry for many users and applications. Zero-knowledge proofs (ZKPs) have emerged as a powerful solution to this challenge.

1.1 Motivation

Elliptic curve pairings represent fundamental building blocks for numerous cryptographic systems including Groth16[?], PLONK[?], BLS signature schemes[?], and KZG polynomial commitment schemes[?].

However, **pairings are computationally intensive**. This creates critical bottlenecks in two important deployment contexts:

1. **Resource-constrained virtual machines** (Ethereum VM, BitVM, ZKVMs): Computation is metered — each operation incurs gas costs. A single pairing verification requiring thousands of field operations becomes economically prohibitive during peak network congestion.
2. **Arithmetized computation systems** (R1CS [?], Plonkish [?], Cairo AIR [?]): Each operation must be represented algebraically, adding significant prover overhead beyond native execution costs.

In such contexts, an efficient proof system for pairing correctness can replace expensive native execution inside the constrained environment.

1.2 Our Contribution

We present a public-coin interactive proof system for verifying pairing computations that consolidates verification into unified polynomial relationships under the Schwartz-Zippel lemma. By treating complete Miller loop iterations as single polynomial relationships rather than sequences of individual extension field operations [?], our approach achieves:

- **35% reduction** in SCS constraints ($1,592,440 \rightarrow 1,035,550$ for a 3-pairing BN254 Groth16 verification check; 33% on BLS12-381) «««< HEAD «««< HEAD
- **53% reduction** in committed elements ($571,125 \rightarrow 269,616$; 54.5% on BLS12-381) =====
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- **53% reduction** in committed elements ($571,125 \rightarrow 269,616$; 54.5% on BLS12-381) »»»> fcab282 (fix: commitment terminology)
- up to **~39% reduction** in compile-time memory and up to **48%** in solver memory

Our key insight is that instead of verifying individual field operations separately, we can verify entire Miller loop iterations as single polynomial identities. This consolidation dramatically reduces both the number of modular operations required and the communication overhead from intermediate variable commitment. «««< HEAD =====
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Relation to prior work. The polynomial identity testing framework of Feltroidprime [?] verifies each individual $\mathbb{F}_{q^{12}}$ multiplication as its own quotient-remainder relation, committing to a remainder for every operation. Our conceptual departure is one of granularity: we consolidate *entire Miller loop iterations*—squaring, residue witness factors, and all line function multiplications of a step—into a single polynomial relation, eliminating the intermediate remainders altogether (Section ??). An early proof-of-concept of these techniques, adapted from a technical write-up and implementation we circulated, was deployed in the Garaga library [?] for Cairo/Starknet in 2024. The purpose of this work is to formally introduce and systematically evaluate the techniques, establishing soundness

87 bounds (Section ??), benchmarking against the state-of-the-art gnark baseline, and gener-
 88 alizing the machinery beyond the Cairo-specific implementation into a reusable toolkit
 89 (Section ??). »»»> fcab282 (fix: commitment terminology)
 90 ===== »»»> 304dec4 (chore: remove repetitive garaga impl ref) As a reusable
 91 artefact of independent interest, we also contribute a **polynomial ring toolkit** for gnark’s
 92 emulated arithmetic package. The toolkit packages the two-round interactive proof system
 93 into a clean API—a single `NewPolyRingCheck` call registers a modulus, `MulPolyRings`
 94 submits multiplications with deferred verification, and `performDeferredRingChecks` closes
 95 the circuit with two `api.Commit` challenges. Atop this, the `PolyRingAccumulator` provides
 96 a higher-level interface for building product chains: factors are enqueued via `Mul`, squarings
 97 via `Sqr`, and the accumulated product is evaluated lazily through `Eval`, with optional
 98 automatic degree-bounding to keep intermediate products manageable.

99 2 Background

Algorithm 1 Optimal Ate pairing over Barreto-Naehrig curves [?]

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2$

Output: $a_{opt}(Q, P)$

```

1: Write  $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$  where  $s_i \in \{-1, 0, 1\}$  and  $s_L = 1$ 
2:  $T \leftarrow Q, f \leftarrow 1$ 
3: for  $i = L - 2$  to  $0$  do
4:    $f \leftarrow f^2 \cdot l_{T,T}(P)$ 
5:    $T \leftarrow [2]T$ 
6:   if  $s_i = 1$  then
7:      $f \leftarrow f \cdot l_{T,Q}(P)$ 
8:      $T \leftarrow T + Q$ 
9:   end if
10:  if  $s_i = -1$  then
11:     $f \leftarrow f \cdot l_{T,-Q}(P)$ 
12:     $T \leftarrow T - Q$ 
13:  end if
14: end for
15:  $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$ 
16:  $f \leftarrow f \cdot l_{T,Q_1}(P), T \leftarrow T + Q_1$ 
17:  $f \leftarrow f \cdot l_{T,-Q_2}(P), T \leftarrow T - Q_2$ 
18:  $f \leftarrow f^{(p^{12}-1)/r}$ 
19: return  $f$ 

```

100 2.1 Pairing Computation

101 An elliptic curve pairing is a non-degenerate bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, where $\mathbb{G}_1$
 102 and $\mathbb{G}_2$ are groups of points on elliptic curves over finite fields, and $\mathbb{G}_T$ is a multiplicative
 103 group in an extension field. For practical cryptographic applications, we require that
 104 this map be efficiently computable. We will focus on the optimal ate pairing [?] over
 105 Barreto-Naehrig (BN) curves, which is widely used due to its efficiency. We will specifically
 106 use the Ethereum’s BN254 curve, defined over a 254-bit prime field $\mathbb{F}_q$, with groups $\mathbb{G}_1, \mathbb{G}_2$
 107 of order r and target group $\mathbb{G}_T$ contained in the 12th extension field $\mathbb{F}_{q^{12}}$.

108 The computation of an optimal ate pairing—the focus of our implementation—consists
 109 of two primary phases:

- 110 1. **Miller loop:** The core iterative computation that evaluates line functions at pairs
 111 of points from $\mathbb{G}_1$ and $\mathbb{G}_2$, accumulating intermediate results to produce a value in
 112 $\mathbb{F}_{q^{12}}$.
- 113 2. **Final Exponentiation:** This step raises the Miller loop output to the power
 114 $(q^{12} - 1)/r$, where r is the order of the groups, ensuring the result lies in the correct
 115 subgroup $\mathbb{G}_T$.

116 While both phases present optimization opportunities, the constraint-based verification
 117 of these computations introduces unique challenges and possibilities that differ significantly
 118 from native implementations.

119 2.2 Schwartz-Zippel lemma

120 The Schwartz-Zippel lemma (or more precisely, the Demillo-Lipton-Schwartz-Zippel
 121 lemma [?, ?, ?, ?]) provides a probabilistic method for testing polynomial identities.
 122 The lemma states that for a non-zero polynomial $P(x_1, x_2, \dots, x_n)$ of total degree d over
 123 a finite field $\mathbb{F}$, when evaluated at uniformly random field elements, the probability that P
 124 evaluates to zero is at most $d/|\mathbb{F}|$.

125 This probabilistic bound becomes negligible for cryptographically sized fields where
 126 $d \ll |\mathbb{F}|$, making it practical to verify polynomial identities by evaluation instead of
 127 coefficient-wise operations.

128 2.3 Computation Models

129 We implement and benchmark our toolkit in gnark [?], the industry-standard Go library
 130 for arithmetized circuit development, targeting both SCS (Sparse Constraint System, the
 131 native Plonkish backend) and R1CS constraint systems. We map each field multiplication
 132 to a constraint (denoted by **C**) to maintain compatibility with the R1CS model used in
 133 established literature. In R1CS [?], prover complexity is dominated by multiplication gates
 134 while additions are free. To synchronize our theoretical analysis with the benchmarks
 135 provided by El Housni [?], we define our cost metric based on field multiplications.

136 2.3.1 R1CS Cost Model

137 We measure computational cost in terms of *constraints* (denoted by **C**), representing the
 138 fundamental unit of work in Rank-1 constraint systems. One constraint corresponds to
 139 one multiplication gate.

140 **Basic Field Operations in $\mathbb{F}_q$:**

- 141 • **Addition/Subtraction:** Free
- 142 • **Multiplication:** One constraint (1C)
- 143 • **Division/Inversion:** One multiplication + one equality check

144 **Commitment Overhead:** Beyond constraint costs, we account for *commitment*
 145 *overhead*—the number of field elements the prover must commit to (via Fiat-Shamir
 146 hashing in non-interactive proofs). For blockchain deployments, this overhead directly
 147 impacts transaction costs and is often the dominant factor.

2.3.2 Cost Model Comparison

Operations that are traditionally expensive (inversions, divisions) become cheap in constraint systems when computed via witnesses. This inverts traditional optimization priorities: we can afford inversions to save multiplications, enabling affine coordinate systems and other techniques impractical in native implementations.

Recognizing and exploiting these unique computational properties is essential for unlocking hidden performance potential. Youssef El Housni’s pioneering work **Pairings in Rank-1 Constraint Systems** [?] represents, to our knowledge, the first systematic exploration of these optimization opportunities, providing a breakthrough in the practicality of recursive proving by thoughtfully leveraging the distinctive cost model inherent to constraint-based systems.

3 Related Work

The landscape of efficient pairing verification in constraint systems has evolved through three key contributions that our work directly builds upon.

1. El Housni’s **Pairings in Rank-1 Constraint Systems** [?] established the theoretical framework for optimizing pairings in arithmetized environments, introducing efficient Miller loop formulas in affine coordinates and sparse line function representations. This work inverts the usual cost model: inversions become cheap (2C) while multiplications remain the dominant cost.
2. Feltroidprime [?] introduced polynomial identity testing for $\mathbb{F}_{q^{12}}$ arithmetic via the Schwartz-Zippel lemma. Each multiplication $A \cdot B$ is expressed as $A(x) \cdot B(x) = Q(x) \cdot P_{12}(x) + R(x)$ and verified at a single random point, with the quotients Q batched via random linear combination. Our work applies the same polynomial framework but at a coarser granularity: entire Miller loop steps rather than individual multiplications, eliminating intermediate remainder commitments entirely. Concretely, a 3-pair zero-bit Miller loop step requires four chained relations and 48 $\mathbb{F}_q$ commitments under [?], versus one relation and 12 commitments in our scheme (Section ??).
3. Novakovic and Eagen’s **On Proving Pairings** [?] showed that final exponentiation can be replaced by a residue witness check, reducing final exponentiation cost by over 85% for BN254. We integrate their residue witness c directly into our polynomial step equations (Section ??), unifying the Miller loop and final exponentiation into the same polynomial verification framework.

The **Garaga** library [?] deployed an early Cairo-specific proof-of-concept of the consolidation technique in 2024, adapted from a preceding technical write-up [?]. Garaga demonstrates practicality in production on Starknet but provides no formal treatment. This paper supplies the missing foundations—the operation polynomials of Section ??, the batched two-round protocol and its soundness analysis in Section ??—and evaluates the technique systematically against the strongest available baseline: gnark’s emulated pairing stack, which incorporates the optimizations of both [?] and [?].

4 Pairing Operations as Polynomials

Algorithm ?? gives the pairing computation combining El Housni’s R1CS optimizations [?] with Novakovic-Eagen’s residue witness technique [?]. The 27th root of unity W scales the Miller loop output f to be a cubic residue (see [?] §4.3); if f is not already a cubic residue, multiplying by W^w , $w \in \{1, 2\}$ corrects this.

Our contribution is to express each step of this algorithm as a single multi-operand polynomial product verified via the Schwartz-Zippel lemma, rather than as a sequence of individual binary multiplications as in [?].

Algorithm 2 Optimal Ate pairing with Residue witness check

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}$

Internal $W, W^2 \in \mathbb{F}_{q^{12}}$

▷ W is 27th root of unity

Output: $a_{opt}(Q, P)$

```

1: Write  $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$  where  $s_i \in \{-1, 0, 1\}$  and  $s_L = 1$ 
2:  $c^{-1} \leftarrow 1/c$                                 ▷ Inverse Residue witness
3:  $f \leftarrow c^{-1}$                                 ▷ Initialize with inverse residue witness
4:  $T \leftarrow Q$                                     ▷  $T_i$  for each pair for multi-pairing
   Every operation involving  $T, P$  or  $Q$  is repeated for  $T_i, P_i$  and  $Q_i$  for multi-pairing.

5: for  $i = L - 2$  to  $0$  do
6:    $f \leftarrow f^2$ 
7:    $f \leftarrow f \cdot l_{T,T}(P)$                     ▷ Repeat for  $T_i, P_i$ 
8:    $T \leftarrow [2]T$                                 ▷ Repeat for  $T_i$ 
9:   if  $s_i = 1$  then
10:     $f \leftarrow f \cdot c^{-1}$                         ▷ Residue witness
11:     $f \leftarrow f \cdot l_{T,Q}(P)$                     ▷ Repeat for  $T_i, Q_i, P_i$ 
12:     $T \leftarrow T + Q$                                 ▷ Repeat for  $T_i, Q_i$ 
13:  else if  $s_i = -1$  then
14:     $f \leftarrow f \cdot c$                                 ▷ Residue witness
15:     $f \leftarrow f \cdot l_{T,-Q}(P)$                     ▷ Repeat for  $T_i, -Q_i, P_i$ 
16:     $T \leftarrow T - Q$                                 ▷ Repeat for  $T_i, -Q_i$ 
17:  end if
18: end for

19:  $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$         ▷ Repeat for  $Q_{1i}, T_i$ 
20:  $f \leftarrow f \cdot l_{T,Q_1}(P)$                     ▷ Repeat for  $Q_{1i}, T_i, P_i$ 
21:  $T \leftarrow T + Q_1$                                 ▷ Repeat for  $Q_{1i}, T_i$ 
22:  $f \leftarrow f \cdot l_{T,-Q_2}(P)$                     ▷ Repeat for  $Q_i, T_i, P_i$ 
23: if  $w = 1$  then
24:    $f \leftarrow f \cdot W$ 
25: else if  $w = 2$  then
26:    $f \leftarrow f \cdot W^2$ 
27: end if
28:  $f \leftarrow f \cdot c^{-q} \cdot c^{q^2} \cdot c^{-q^3}$         ▷ Uses Frobenius maps, Negative exponents use  $c^{-1}$ 

29: return  $f$ 

```

4.1 Comparative Analysis

We compare the two schemes for a 3-pair Miller loop zero-bit step (the most common case in BN254's NAF[?] representation of $s = 6t + 2$). Here $F \in \mathbb{F}_{q^{12}}$ is the accumulator, $L_i \in \text{Sparse}_{01379}$ are the three doubling line functions, $R \in \mathbb{F}_{q^{12}}$ is the updated accumulator, and Q is the quotient polynomial. Sparse_{01379} denotes the subspace of $\mathbb{F}_{q^{12}}$ elements whose only nonzero coefficients are at degrees 0, 1, 3, 7, 9—the form that BN254 line functions naturally take [?], reducing their evaluation cost from 12C to 4C. All compared schemes—[?], [?], and ours—exploit this sparseness of the Miller line functions; the corresponding

savings are included in every constraint count below.

4.1.1 Previous scheme

Adapting [?] to lines ??-?? of Algorithm ?? for a 3-pair Miller loop gives:

$$F(x) \cdot F(x) \stackrel{?}{=} Q_1(x) \cdot P_{12}(x) + T_1(x)$$

$$T_1(x) \cdot L_1(x) \stackrel{?}{=} Q_2(x) \cdot P_{12}(x) + T_2(x)$$

$$T_2(x) \cdot L_2(x) \stackrel{?}{=} Q_3(x) \cdot P_{12}(x) + T_3(x)$$

$$T_3(x) \cdot L_3(x) \stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x)$$

Evaluating F , T_1 – T_3 , R costs $12C$ each and L_1 – L_3 cost $4C$ each; all four remainder polynomials require commitment. **Total: 76C and 48 $\mathbb{F}_q$ commitments** (4×12).

4.1.2 Our scheme

Our scheme consolidates lines ??-?? into a single polynomial identity:

$$F(x) \cdot F(x) \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) \stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x)$$

By eliminating intermediate T_* polynomials, only R requires commitment. **Total: 41C and 12 $\mathbb{F}_q$ commitments** ($2 \times 12C + 3 \times 4C + 5C$).

This results in **46% fewer constraints** and a **75% reduction in commitments**. When verifying ZK proofs on-chain, commitment savings are often more critical than constraint savings, due to the calldata costs associated with transmitting polynomial hints.

4.1.3 Performance Comparison

Table ?? compares the computational costs and commitment overhead between the schemes for a single 0-bit Miller loop operation with 3 pairs.

Table 1: 3-Pair Miller loop 0-bit constraints comparison

Scheme	Constraints	$\mathbb{F}_q$ Commitments	Improvement
[?]	132C	—	Baseline
[?]	76C*	48	42%
Our scheme	41C*	12	69%
[?]: 1 square ($36C$) + 3 sparse mul ($3 \times 30C$) = $132C$ [?]: 5 evals ($5 \times 12C$) + 3 line evals ($3 \times 4C$) + 4 muls ($4 \times 1C$) = $76C$ Our scheme: 2 evals ($2 \times 12C$) + 3 line evals ($3 \times 4C$) + 5C (muls + RLC) = $41C$ *Costs exclude Fiat-Shamir commitment overhead			

4.2 Operation Polynomials

Our implementation uses different polynomials for different bits in the Miller loop. Let us walk through a 3-pairing setup, the configuration used for Groth16 verification (Section ??). We will describe the polynomials for zero bits, non-zero bits and correction step.

4.2.1 Miller loop: Zero Bit Equation

As seen in ?? for zero bit operations (line ?? and ?? with three $l_{T,T}(P)$ lines in Algorithm ??), the verification equation is:

$$F^2(x) \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (1)$$

Table 2: Zero bit operation polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	$11 \times 2 = 22$	12
$L_1, L_2, L_3 \in \text{Sparse}_{01379}$	The line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total constraints	36C Evals, 4C Muls and 1C RLC		41
Q (<i>Amortized See ??</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	38	39

4.2.2 Miller loop: Non-Zero Bit Equation

For non-zero bit operations (lines ??, ??, ??, ?? with three pairs of lines in Algorithm ??), the Miller loop performs both doubling and addition steps. So for each pair we have two Sparse_{01379} lines. The verification equation becomes:

$$F^2(x) \cdot W(x) \cdot L_{d1}(x) \cdot L_{d2}(x) \cdot L_{d3}(x) \cdot L_{a1}(x) \cdot L_{a2}(x) \cdot L_{a3}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (2)$$

Table 3: Non-zero bit operation polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	$11 \times 2 = 22$	12
$L_{d1}, L_{d2}, L_{d3} \in \text{Sparse}_{01379}$	Doubling line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
$L_{a1}, L_{a2}, L_{a3} \in \text{Sparse}_{01379}$	Addition line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
W or $W^{-1} \in \mathbb{F}_{q^{12}}$	Residue witness	11	12
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total constraints	60C Evals, 8C Muls and 1C RLC		69
Q (<i>Amortized See ??</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	76	77

The residue witness W encapsulates the equivalence class relationship established by Novakovic and Eagen's final exponentiation elimination. The inverse of W is used for negative bits, this just changes the coefficients the equation remains the same.

4.2.3 Miller loop: Frobenius mapping Equation

The correction step finalizes the Miller loop by incorporating the Frobenius endomorphism corrections (lines ?? and ?? in Algorithm ??). Again, we have two Sparse_{01379} lines for each pair, for π_p and $-\pi_p^2$ mappings, without any squaring of the accumulator. The verification equation becomes:

$$F(x) \cdot L_{1\pi}(x) \cdot L_{2\pi}(x) \cdot L_{3\pi}(x) \cdot L_{1\pi^2}(x) \cdot L_{2\pi^2}(x) \cdot L_{3\pi^2}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (3)$$

Table 4: Correction step polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	11	12
$L_{1\pi}, L_{2\pi}, L_{3\pi} \in \text{Sparse}_{01379}$	π_p correction lines	$9 \times 3 = 27$	$4 \times 3 = 12$
$L_{1\pi^2}, L_{2\pi^2}, L_{3\pi^2} \in \text{Sparse}_{01379}$	$-\pi_p^2$ correction lines	$9 \times 3 = 27$	$4 \times 3 = 12$
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total constraints	48C Evals, 6C Muls and 1C RLC		55
Q (<i>Amortized See ??</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	54	55

4.2.4 Final Exponentiation Equation

The final exponentiation check is handled as described in [?]. The equation handles the cubic scale factor multiplication, and **Frobenius component** ($q - q^2 + q^3$) of the final exponentiation (lines ??, ?? and ?? in Algorithm ??). The verification equation looks like:

$$F(x) \cdot c^{-q}(x) \cdot c^{q^2}(x) \cdot c^{-q^3}(x) \cdot W^w(x) = 1 + Q(x) \cdot P_{12}(x) \quad (4)$$

where $w \in \{0, 1, 2\}$ determines the cubic root of unity scaling factor, and the Frobenius mappings are computed using the residue witness c and its inverse and provided as input. Cubic root is sparse element with just 2 coefficients the degree ranging from 8 to 10 because of positioning of the coefficients. Our R is now 1 for a successful proof verification.

Table 5: Final exponentiation polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Final Miller loop result	11	12
$c^{-q} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius $-q$	11	12
$c^{q^2} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius q^2	11	12
$c^{-q^3} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius $-q^3$	11	12
$W^w \in \mathbb{F}_{q^{12}}$	Cubic scale factor	10	2
$R = 1$	Unity (constant)	0	1
Total constraints	51C Evals, 4C Muls and 1C RLC		56
Q (<i>Amortized See ??</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	43	44

With this we have all the polynomials we need for verifying entire pairing operation. Next we will look into high level overview of pairing verification via an interactive oracle proof.

4.3 Security

Each verification equation takes the form $P = \prod_i A_i(x) - Q(x) \cdot P_{12}(x) - R(x)$ where $\deg(P) < 2^8$.¹ By the Schwartz-Zippel lemma (Section ??):

Soundness: $\delta_s \leq d/|\mathbb{F}_q| < 2^8/2^{254} = 2^{-246}$ for BN254.

Completeness: An honest prover always produces $P(x) = 0$ identically, so the verifier always accepts. Euclidean division guarantees unique Q, R with $\deg(R) < \deg(P_{12}) = 12$.

5 Full Pairing Proof with Quotient Batching

We now combine the polynomial equations from Section ?? into a complete two-round public-coin interactive proof, further reducing verification cost by batching all quotients into a single accumulated polynomial.

¹For k -pair pairings, $\deg(P) \approx 88 + 18(k - 3)$, well below 2^8 for all practical k .

5.1 Batching Polynomial Identity Testing

All n polynomial equations from Section ?? are batched into a single verification. The prover commits to remainders $\{R_i\}_{i=0}^{n-1}$ and receives challenge z . Using z , the prover forms the accumulated quotient $Q_{\text{acc}} = \sum_{i=0}^{n-1} z^i \cdot Q_i$. The verifier then samples a second challenge point c and checks:

$$\sum_{i=0}^{n-1} z^i \cdot [A_i(c) \cdot B_i(c) \cdots X_i(c) - R_i(c)] = Q_{\text{acc}}(c) \cdot P_{12}(c) \quad (5)$$

This reduces n polynomial verifications (degrees 38–76, see Tables ??–??) to a single evaluation of a degree-76 polynomial.

5.2 Prover's Algorithms

The prover computes quotients Q_i and remainders R_i for each equation in Section ?. For each step, it calls POLYMULDIV, which multiplies the input polynomials and performs standard Euclidean division by the irreducible P_{12} . After receiving challenge z , the prover forms $Q_{\text{acc}} = \sum_i z^i Q_i$ and sends $\mathcal{R} = \{R_i\}$ and Q_{acc} to the verifier. The full prover procedure is given in Algorithm ??

5.3 Verifier's Algorithms

The verifier receives $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$, residue witness c [?], the W (27th root of unity), the remainder array $\mathcal{R}$, and Q_{acc} from the prover. Using challenge z from the first round and a freshly sampled x , it mirrors the prover's pairing loop while evaluating each polynomial at x , accumulating the RLC e_{acc} . The final check is Equation ?. The full verifier is given in Algorithm ??

Algorithm 3 Pairing Prover: Preparing polynomials

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}$
 Internal: $W, W^2 \in \mathbb{F}_{q^{12}}$ $\triangleright W$ is 27th root of unity

Output: Arrays $\mathcal{Q}, \mathcal{R}$ of quotients and remainders
 Write $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$

- 1: $c^{-1} \leftarrow 1/c$ $\triangleright$ Inverse Residue witness
- 2: $f \leftarrow c^{-1}$ $\triangleright$ Initialize with inverse residue witness
- 3: $T \leftarrow Q$ $\triangleright T_i$ for each pair for multi-pairing

Every operation involving T, P or Q is repeated for T_i, P_i and Q_i for multi-pairing.

4: **function** PolyMulDiv ($R, P_1, \dots, P_n \in \mathbb{F}_q[x]$): **return** $Q, R \in \mathbb{F}_q[x]$
 5: s.t. $\prod P_i = Q \cdot P_{12} + R, \deg(R) < \deg(P_{12})$

Miller loop

- 6: **for** $i = L - 2$ to 0 **do**
- 7: $L_{\text{dbl}} \leftarrow l_{T,T}(P), T \leftarrow [2]T$
- 8: **if** $s_i = 0$ **then**
- 9: $Q, R \leftarrow \text{PolyMulDiv}(f, f, L_{\text{dbl}})$ $\triangleright$ add L_{*i} for each P_i, Q_i
- 10: **else if** $s_i = 1$ **then**
- 11: $L_{\text{add}} \leftarrow l_{T,Q}(P), T \leftarrow T + Q$
- 12: $Q, R \leftarrow \text{PolyMulDiv}(f, f, c^{-1}, L_{\text{dbl}}, L_{\text{add}})$ $\triangleright$ add L_{*i} for each P_i, Q_i
- 13: **else if** $s_i = -1$ **then**
- 14: $L_{\text{add}} \leftarrow l_{T,-Q}(P), T \leftarrow T - Q$
- 15: $Q, R \leftarrow \text{PolyMulDiv}(f, f, c, L_{\text{dbl}}, L_{\text{add}})$ $\triangleright$ add L_{*i} for each P_i, Q_i
- 16: **end if**
- 17: $f \leftarrow R, \mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$
- 18: **end for**

- 19: $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$
- 20: $L_\pi \leftarrow l_{T,Q_1}(P)$
- 21: $T \leftarrow T + Q_1$
- 22: $L_{-\pi^2} \leftarrow l_{T,-Q_2}(P)$
- 23: $Q, R \leftarrow \text{PolyMulDiv}(f, L_\pi, L_{-\pi^2})$ $\triangleright$ add L_{*i} for each P_i, Q_i
- 24: $f \leftarrow R, \mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$

Final exponentiation: Cubic scaling + Frobenius mappings

- 25: **if** $w = 0$ **then**
- 26: $Q, R \leftarrow \text{PolyMulDiv}(f, c^{-q}, c^{q^2}, c^{-q^3})$
- 27: **else if** $w = 1$ **then**
- 28: $Q, R \leftarrow \text{PolyMulDiv}(f, W, c^{-q}, c^{q^2}, c^{-q^3})$
- 29: **else if** $w = 2$ **then**
- 30: $Q, R \leftarrow \text{PolyMulDiv}(f, W^2, c^{-q}, c^{q^2}, c^{-q^3})$
- 31: **end if**
- 32: $\mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$ $\triangleright$ Final $\mathcal{Q}$ and $\mathcal{R}$ accumulation, skip f

33: **return** $\mathcal{Q}, \mathcal{R}$

Algorithm 4 Pairing Verifier: Evaluating remainders

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}, \mathcal{R}$ array of remainders, $Q_{\text{acc}} \in \mathbb{F}_q[x]$

 Internal: 27th root of unity, $W, W^2 \in \mathbb{F}_{q^{12}}$, first round challenge, $z \xleftarrow{\$} \mathbb{F}_q$
Output: $\text{pairing}(P, Q) \stackrel{?}{=} 1$

 Write $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$

```

1:  $c^{-1} \leftarrow 1/c$  ▷ Inverse Residue witness
2:  $f \leftarrow c^{-1}$  ▷ Initialize with inverse residue witness
3:  $T \leftarrow Q$  ▷  $T_i$  for each pair for multi-pairing
4:  $x \xleftarrow{\$} \mathbb{F}_q$  ▷ Sample evaluation point from field
5:  $e_{\text{acc}} \leftarrow 0, a_{\text{rlc}} \leftarrow 1$  ▷ Evaluation accumulator and random linear combination multiplier
  Every operation involving  $T, P$  or  $Q$  is repeated for  $T_i, P_i$  and  $Q_i$  for multi-pairing.

```

 6: **function** EvalPoly ($R, P_1, \dots, P_n \in \mathbb{F}_q[x]$): **return** $(\prod_{i=1}^n P_i(z) - R(z) \in \mathbb{F}_q)$

Miller loop

```

7: for  $i = L - 2$  to  $0$  do
8:    $L_{\text{dbl}} \leftarrow l_{T,T}(P), T \leftarrow [2]T$ 
9:    $R \leftarrow \mathcal{R}.\text{pop\_front}()$  ▷ Fetch R from prover
10:  if  $s_i = 0$  then
11:     $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, L_{\text{dbl}})$  ▷ add  $L_{*i}$  for each  $P_i, Q_i$ 
12:  else if  $s_i = 1$  then
13:     $L_{\text{add}} \leftarrow l_{T,Q}(P), T \leftarrow T + Q$ 
14:     $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, c^{-1}, L_{\text{dbl}}, L_{\text{add}})$  ▷ add  $L_{*i}$  for each  $P_i, Q_i$ 
15:  else if  $s_i = -1$  then
16:     $L_{\text{add}} \leftarrow l_{T,-Q}(P), T \leftarrow T - Q$ 
17:     $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, c, L_{\text{dbl}}, L_{\text{add}})$  ▷ add  $L_{*i}$  for each  $P_i, Q_i$ 
18:  end if
19:   $f \leftarrow R, a_{\text{rlc}} \leftarrow a_{\text{rlc}} \cdot z$  ▷ Update random linear combination factor
20: end for

21:  $R \leftarrow \mathcal{R}.\text{pop\_front}()$  ▷ Fetch R from prover
22:  $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$ 
23:  $L_\pi \leftarrow l_{T,Q_1}(P)$ 
24:  $T \leftarrow T + Q_1$ 
25:  $L_{-\pi^2} \leftarrow l_{T,-Q_2}(P)$ 
26:  $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, L_\pi, L_{-\pi^2})$  ▷ add  $L_{*i}$  for each  $P_i, Q_i$ 
27:  $f \leftarrow R, a_{\text{rlc}} \leftarrow a_{\text{rlc}} \cdot z$  ▷ Update random linear combination factor

```

Final exponentiation: Cubic scaling + Frobenius mappings

```

28:  $R \leftarrow \mathcal{R}.\text{pop\_front}()$  ▷ Fetch R from prover
29: if  $w = 0$  then
30:    $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, c^{-q}, c^{q^2}, c^{-q^3})$ 
31: else if  $w = 1$  then
32:    $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, W, c^{-q}, c^{q^2}, c^{-q^3})$ 
33: else if  $w = 2$  then
34:    $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, W^2, c^{-q}, c^{q^2}, c^{-q^3})$ 
35: end if

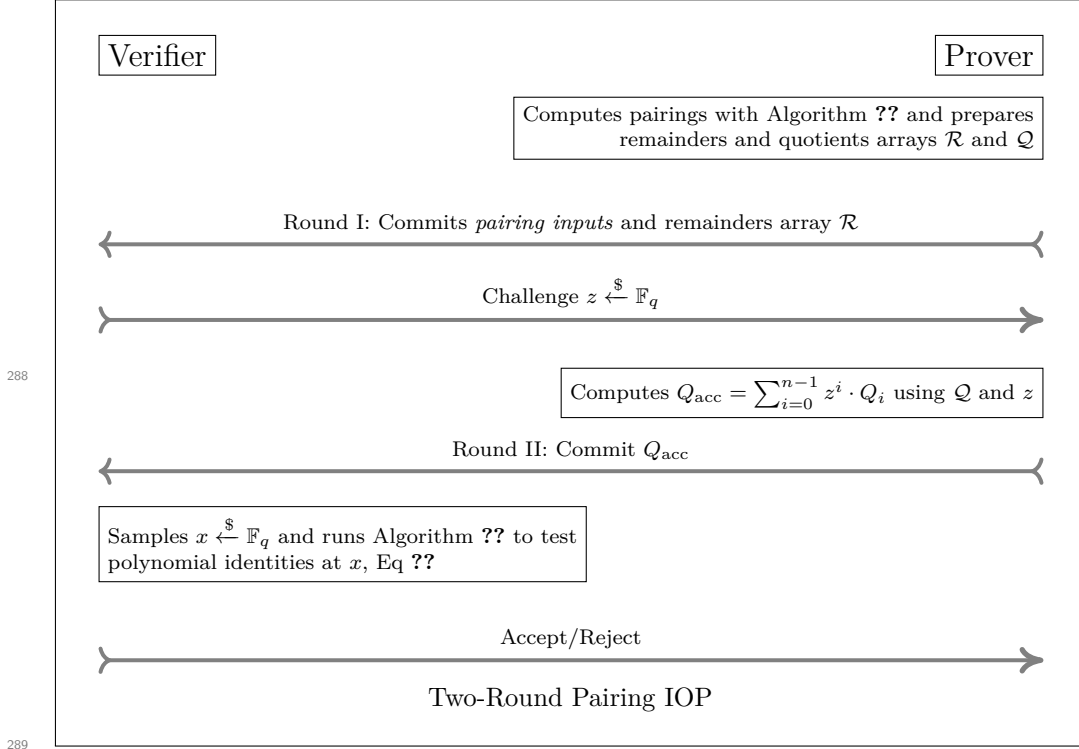
```

```

36: return  $Q_{\text{acc}}(x) \stackrel{?}{=} e_{\text{acc}} \wedge R \stackrel{?}{=} 1$  ▷ Polynomial identity and expected pairing output check

```

5.4 Interactive Proof



5.5 Fiat-Shamir Transformation

Applying the Fiat-Shamir transformation [?] with random oracle H to the two-round protocol of Section ?? gives a non-interactive proof with knowledge soundness $\delta_s + 3(m^2 + 1)2^{-\lambda}$ [?], where $m = 2$ is the number of oracle queries (one per round) and λ is the challenge bit-length; Section ?? instantiates this bound.

5.5.1 Non-Interactive Proof

The prover runs Algorithm ?? to obtain $(\mathcal{Q}, \mathcal{R})$, derives the challenge

$$z = H(\text{pairing inputs}, c, W^w, \mathcal{R})$$

from the transcript instead of receiving it, computes $Q_{\text{acc}} = \sum_{i=0}^{n-1} z^i \cdot Q_i$, and outputs the proof $(\text{pairing inputs}, c, W^w, \mathcal{R}, Q_{\text{acc}})$. The verifier recomputes z , derives the evaluation point $x = H(z, Q_{\text{acc}})$, and runs Algorithm ??, testing the polynomial identities via Equation (??).

The soundness bound of Theorem ?? carries over because every prover-chosen value entering Equation (??) is absorbed before the challenge it must not depend on: $\mathcal{R}$ and the hinted witnesses c, W^w before z , and Q_{acc} with z chained into the second query before x .

5.5.2 Instantiating the Random Oracle

The protocol is agnostic to how H is realised; the cost of absorbing one $\mathbb{F}_q$ element into the transcript varies across execution environments. We therefore qualify the committed element counts of Section ?? and Table ?? for our two instantiations.

CairoVM (Poseidon). Prover-supplied hints are passed as transaction calldata and absorbed into a running Poseidon [?] sponge; $\mathbb{F}_q$ elements are represented by 96-bit limbs, with two limbs packed into each `Felt252` absorption. Both hashing cost and calldata therefore scale linearly with the number of committed elements, which is why the COMMITTS column of Table ?? is reported alongside modular operations: halving the committed elements roughly halves the transcript-related share of VM steps and the hint calldata.

Faster Hash Verification [?] in gnark. Belling et al. [?] provide commitment schemes that allow the prover to compute commitment natively (outside the circuit) and attach an Argument of Knowledge to prove correctness. The scheme is described for both Groth16 and PLONK ([?] Section 6.1 and 6.2). Belling later extends this to support multiple commitment for multiple rounds for gnark in [?].

This is made available via `api.Commit` calls in gnark. The costs are backend-specific.

Groth16 in gnark. Each commitment costs two $\mathbb{G}_1$ MSMs of size equal to the number of committed wires ([?] Section 6.1). Proof size grows by one $\mathbb{G}_1$ point per commitment, plus a single knowledge proof folded across all commitments. Verification gains one two-pairing product check (the knowledge proof) and one extra scalar multiplication in the public-input MSM per commitment. None of this touches the constraint count, since the commitment lives entirely outside the arithmetization.

PLONK in gnark. The overhead shifts almost entirely into the constraint system instead. Each committed wire adds one constraint ([?] Section 6.2), so cost scales with the number of committed *wires*. Proof size grows by one $\mathbb{G}_1$ element per commitment. Verification pays no extra pairing: the committed polynomial's evaluation is folded into the batched KZG opening that PLONK already performs, so the marginal verifier cost is a handful of field operations.

5.6 Security Analysis

Throughout this section, $z, x \in \mathbb{F}_q$ denote field element challenges (scalars), while polynomials are elements of $\mathbb{F}_q[X]$ with X as the formal indeterminate. In particular, z^j is a scalar coefficient, not a polynomial term.

5.6.1 Algebraic Intuition

Before the formal soundness proof it is instructive to consider why an incorrect remainder is difficult to hide. Suppose a malicious prover corrupts exactly one remainder: $R'_j = R_j + \Delta_j$ with $\Delta_j \in \mathbb{F}_q[X]$, $\Delta_j \neq 0$, $\deg(\Delta_j) \leq 11$. Since z is obtained *after* the commitment to $\mathcal{R}$, the prover cannot adjust any other R_k to compensate for Δ_j . To successfully deceive the verifier, the prover must find a valid forged quotient $Q'_{\text{acc}} \in \mathbb{F}_q[X]$ satisfying

$$\sum_{i=0}^{n-1} z^i (A_i - R'_i) = Q'_{\text{acc}} \cdot P_{12}$$

All $R'_i = R_i$ except for $R'_j = R_j + \Delta_j$. Substituting $R'_j = R_j + \Delta_j$ gives

$$Q'_{\text{acc}} \cdot P_{12} = Q_{\text{acc}} \cdot P_{12} - z^j \Delta_j(x)$$

$$Q'_{\text{acc}} = Q_{\text{acc}} - \frac{z^j \Delta_j(x)}{P_{12}(x)}.$$

Here $z^j \in \mathbb{F}_q$ is a nonzero scalar (with overwhelming probability over the choice of z) and $\Delta_j(x) \in \mathbb{F}_q[X]$ is a nonzero polynomial of degree at most 11. Since P_{12} is irreducible of degree 12 and $\deg(z^j \Delta_j) \leq 11 < 12 = \deg(P_{12})$, the polynomial P_{12} does not divide $z^j \Delta_j$ in $\mathbb{F}_q[X]$; hence $z^j \Delta_j / P_{12}$ is not a polynomial. Consequently no valid $Q'_{\text{acc}} \in \mathbb{F}_q[X]$ can satisfy the equation for this single-corruption case.

The same degree argument extends immediately to an adversary who corrupts *multiple* remainders: for any scalar z , the weighted sum $\sum_i z^i \Delta_i(x)$ is still a polynomial in x of degree at most 11 (since each z^i is a scalar and each $\deg(\Delta_i) \leq 11$), so it cannot be divisible by P_{12} of degree 12 unless it is the zero polynomial, which requires every $\Delta_i = 0$. The formal soundness proof below re-derives this uniformly via Schwartz-Zippel.

5.6.2 Soundness

Theorem 1 (Soundness). *Let n be the number of polynomial equations batched via a random linear combination into the bivariate polynomial identity:*

$$P_{\text{final}}(z, x) = \sum_{i=0}^{n-1} z^i \cdot \left[\prod_j P_{i,j}(x) - R_i(x) \right] - Q_{\text{acc}}(x) \cdot P_{12}(x) = 0$$

Let the maximum degree in x of the polynomial products be $d_x < 2^8$. For a prover committing to incorrect remainders, the probability of successful forgery δ_s over randomly chosen $(z, x) \in \mathbb{F}_q^2$ is bounded by:

$$\delta_s \leq \frac{n - 1 + d_x}{|\mathbb{F}_q|}$$

Proof. If the prover commits to an incorrect remainder $R_i(x)$, the evaluation yields a non-zero error polynomial $P_{\text{err}}(z, x)$. The structure of $P_{\text{final}}(z, x)$ establishes the following degree bounds for $P_{\text{err}}(z, x)$:

- **Degree in z :** at most $n - 1$ (due to the z^i weighting).
- **Degree in x :** at most $d_x < 2^8$ (due to the polynomial products).

By the Schwartz-Zippel lemma, the probability that the non-zero polynomial P_{err} evaluates to zero is bounded by the total degree divided by the field size:

$$\Pr[P_{\text{err}}(z, x) = 0] \leq \frac{\deg_z(P_{\text{err}}) + \deg_x(P_{\text{err}})}{|\mathbb{F}_q|} \leq \frac{n - 1 + d_x}{|\mathbb{F}_q|}$$

□

Concrete parameters for BN254: With $n = 67$ equations and $d_x \leq 256$:

$$\delta_s \leq \frac{66 + 256}{2^{254}} < \frac{2^9}{2^{254}} = 2^{-245}.$$

Non-interactive proof via Fiat-Shamir : From Section ??, under the random oracle model the knowledge soundness error becomes [?]:

$$\delta'_s = \delta_s + 3(m^2 + 1) \cdot 2^{-\lambda}$$

where $m = 2$ (one query per round) and $\lambda = 254$ (challenge bit-length), giving

$$\delta'_s \leq 2^{-245} + 15 \cdot 2^{-254} < 2^{-244}.$$

362 *Remark 1* (Correctness of polynomial evaluation). Theorem ?? assumes that the verifier
 363 evaluates all polynomials at the challenge point x without arithmetic error. The circuit
 364 implementation uses the deferred-reduction inner-product optimisation of Section ??
 365 (Section ??), which accumulates limb-wise products over $\mathbb{Z}$ before a single final reduction.
 366 Correctness of this optimisation requires that no intermediate limb value reaches the
 367 native field characteristic p . We verify this bound concretely for BN254 in Lemma ??
 368 (Section ??).

369 5.6.3 Completeness

370 For an honest prover executing Algorithm ?? with valid inputs:

1. **Polynomial construction:** At each operation, the prover computes following polynomials:

$$\prod_j P_{i,j} = Q_i \cdot P_{12} + R_i$$

2. **RLC formation:** Given challenge z , the prover forms:

$$Q_{\text{acc}} = \sum_{i=0}^{n-1} z^i \cdot Q_i$$

- 371 3. **Verifier's check:** The verifier evaluates at point x :

$$\begin{aligned} \text{LHS} &= \sum_{i=0}^{n-1} z^i \cdot \left[\prod_j P_{i,j}(x) - R_i(x) \right] \\ &= \sum_{i=0}^{n-1} z^i \cdot Q_i(x) \cdot P_{12}(x) \quad (\text{by construction}) \\ &= P_{12}(x) \cdot \sum_{i=0}^{n-1} z^i \cdot Q_i(x) \\ &= P_{12}(x) \cdot Q_{\text{acc}}(x) = \text{RHS} \end{aligned}$$

376 Since the equality holds for any evaluation point x , the protocol satisfies perfect
 377 completeness.

378 6 Polynomial Ring Toolkit

379 The protocol of Section ?? is packaged as a reusable toolkit in the `emulated` package
 380 of gnark [?], parameterised by any emulated field $\mathbb{F}_q$ (via `FieldParams` type T) and any
 381 modulus polynomial $P \in \mathbb{F}_q[x]$. The toolkit operates in a recursive proof setting where
 382 the prover's computations are performed outside the circuit via `hints` (Section ??), while
 383 the verifier's checks are implemented as constraints inside the circuit.

384 Refer to our fork at <https://anonymous.4open.science/r/tiny-gnark-DDCD> for
 385 code.

386 6.1 Ring Representation

387 A ring element is essentially a coefficient slice `[]*Element[T]` in ascending degree order. A
 388 `nil` entry is treated as zero by `InnerProductNoReduce`—the term is skipped entirely—so
 389 sparse polynomials carry no wasted constraint variables.

390 The optional `modEvalFn` of type `EvalFnType[T] = ([*Element[T]]) → *Element[T]`
 391 allows handling small-integer coefficients efficiently by multiplying by the small positive
 392 integer first and negating afterwards, rather than multiplying by the full-width field
 393 representative $q - c$.

394 **Example: BN254 $\mathbb{F}_{q^{12}}$**

395 $P_{12}(x) = x^{12} - 18x^6 + 82$ has nine zero coefficients (stored as `nil`) and two small non-zero
 396 ones. The `modEvalFn` computes $P_{12}(\alpha) = \text{MulConst}(\alpha^0, 82) + \text{Neg}(\text{MulConst}(\alpha^6, 18)) + \alpha^{12}$,
 397 keeping intermediate values small instead of multiplying by $q - 18$.

398 6.2 Hint Functions

399 Hints allow defining computations outside of a circuit. A hint defines an annotated
 400 function in the circuit at compile time. Inputs and outputs are injected at the solving
 401 time. We define these two hints to implement the prover's steps in the two-round protocol
 402 of Section ??.

403 **polyRingMulHint** implements `POLYMULDIV`: it multiplies the input polynomials,
 404 divides by P to obtain (Q, R) .

405 **quotientsRLCHint** computes the accumulated quotient $Q_{\text{acc}} = \sum_i z^i \cdot Q_i \bmod q$
 406 coefficient-wise, corresponding to the prover's step in Algorithm ??.

407 6.3 Algorithms and Implementation

408 6.3.1 API Reference

409 Five functions cover the full lifecycle of a ring group: registration, per-operation multi-
 410 plication, chained accumulation, extra commitment inputs, and finalisation. The three
 411 subsections below trace how a concrete circuit, BN254 pairing verification, calls these: one
 412 shared skeleton, prover work running inline within it, verifier work deferred to its end.

413 **NewPolyRingCheck(mod, modEvalFn)** Registers a *ring group*: a modulus polyno-
 414 mial P paired with an optional small-coefficient evaluator `modEvalFn`. Returns a
 415 `*PolyRingGroupChecks` handle shared by every multiplication under that modulus;
 416 multiple groups with different moduli may coexist in the same circuit.

417 **MulPolyRings(group, $A_1, \dots, A_k$)** Computes $R = (\prod_j A_j) \bmod P$ outside the circuit
 418 via `polyRingMulHint` (prover), range-checks R 's limbs inside the circuit (verifier), and
 419 enqueues the tuple $(A_j; R; Q)$ in the group's deferred check list. Returns R immediately
 420 so callers can continue building the circuit.

421 **PolyRingAccumulator** A helper that accumulates a product chain $\prod_i F_i$ and flushes to
 422 `MulPolyRings` automatically when the accumulated degree would exceed a configurable
 423 bound `targetDeg`, avoiding excessive hint call overhead. `Mul` enqueues one factor; `Sqr`
 424 enqueues the current state twice; `Eval` forces an immediate flush and returns the remainder.

425 **ToCommit($e_1, \dots$)** Adds extra elements to the first commitment (the one that derives chal-
 426 lenge z). Required for hinted values—such as the residue witness c in pairing checks—whose
 427 correctness is asserted *inside* the polynomial identity protocol itself, so they must be ab-
 428 sorbed before z is sampled.

429 **performDeferredRingChecks (internal)** Called once by the gnark backend at circuit
 430 finalisation. Implements Equation (??) via two sequential `api.Commit` calls (Section ??),
 431 producing z then x , followed by one `AssertIsEqual` per group. No in-circuit multiplications
 432 over ring elements occur before this point beyond the remainder range-checks.

433 6.3.2 Algorithmic overview

434 Files `sw_bn254/pairing.go` and `sw_bls12381/pairing.go` implement the Miller loop
 435 and final exponentiation polynomial relations for BN254 and BLS12–381. respectively,
 436 using the polynomial ring toolkit. The prover’s work is entirely in hint functions called
 437 during witness generation. Each `MulPolyRings` call triggers `polyRingMulHint`, which multiplies the input
 438 polynomials and performs Euclidean division by P , returning (Q_i, R_i) . The remainder R_i
 439 is returned to the caller immediately; the quotient Q_i is stored in the group’s deferred
 440 list. In the BN254 pairing (`sw_bn254/pairing.go`), the Miller loop drives this through
 441 a `PolyRingAccumulator` with `targetDeg=69`: `Sqr` and `Mul` accumulate line evaluations
 442 until the degree bound is reached, at which point `Eval` flushes to `MulPolyRings`. The
 443 residue witness c and its inverse are added to the first commitment via `ToCommit` so that
 444 z binds to them before the identity check.

446 6.3.3 Verifier path

447 All verifier work is deferred to `performDeferredRingChecks`, which runs once at circuit
 448 finalisation. It issues two sequential `api.Commit` calls (Section ??): the first absorbs all
 449 remainder limbs (and any `ToCommit` elements), producing z ; the second absorbs z and the
 450 limbs of Q_{acc} (computed by `quotientsRLCHint`), producing x . Chaining z into the second
 451 commitment realises the non-interactive transcript of Section ?. Both challenges are
 452 native field elements; `NativeToEmulated` decomposes each into $\lceil \lambda/b \rceil$ emulated limbs and
 453 asserts the recomposition in-circuit. Polynomial evaluations at x are then computed by
 454 `evalPolyWithChallenge`, which caches the power vector $\mathcal{X} = (1, x, x^2, \dots)$ and evaluates
 455 each polynomial as an inner product via `InnerProductNoReduce` (Section ??); each
 456 polynomial’s evaluation is cached on first use so shared factors (e.g. the modulus P) are
 457 not re-evaluated.

458 6.4 Usage

```
459 // Register the ring group once at initialisation
460 fp, _ := emulated.NewField[emulated.BN254Fp](api)
461 modPoly := fp.MakePoly(82, 0, 0, 0, 0, 0, -18, 0, 0, 0, 0, 0, 1)
462 group := fp.NewPolyRingCheck(modPoly, nil)
463 // Operations on ring elements A, B, C
464 res, _ = fp.MulPolyRings(group, A, B, C) // compute (A·B·C)
465 // Accumulate a product chain
466 acc := fp.NewPolyRingAccumulator(group, targetDeg)
467 acc.Mul(A).Mul(B).Sqr() // enqueue (A·B)2 without multiplying yet
468 rem := acc.Eval()      // multiply and return the remainder
```

469 6.5 Implementation Details and Performance

470 6.5.1 Schoolbook vs Horner’s Method for Polynomial Evaluation

Horner’s method for evaluations in `EvalPoly` across emulated elements causes a linear increase in the limbs size and count per coefficient. This necessitates reductions after each

multiplication. Instead, we cache powers of the evaluation point x in $\mathcal{X} = (x^0, x^1, \dots, x^d)$ and evaluate each polynomial as an inner product of $\mathcal{X}$ and the coefficient array $\mathbf{F}$:

$$\mathcal{X} \cdot \mathbf{F} = \mathcal{X}_0 \cdot F_0 + \mathcal{X}_1 \cdot F_1 + \dots + \mathcal{X}_d \cdot F_d$$

In the emulated field context, Horner’s method evaluations are replaced by this inner product with $\mathcal{X}$. This allows us to use the inner product optimization described below.

6.5.2 Native field operations for inner products

Modular reductions can be deferred or entirely elided for operations expressible as inner products, such as polynomial evaluations and random linear combinations. This optimization leverages the capacity headroom provided by representing small emulated limbs (e.g., 64-bit limbs) within a significantly larger native field element.

Deferred Reduction: Intermediate scalars are permitted to grow over the integers $\mathbb{Z}$ during limb-wise multiplications and linear accumulations.

Preservation of Identity: Algebraic identities are strictly preserved provided the maximum accumulated limb magnitude remains bounded below the native characteristic p .

Handling Asymmetry: In circuits with asymmetric multiplicative depths, operands may accumulate as disparate polynomial multiples of the emulated characteristic p_{emul} . Consequently, a terminal single-element reduction suffices for equivalence testing, bypassing the constraint overhead of intermediate normalization.

Lemma 1 (Overflow safety for deferred reduction). *Let $\mathbb{F}_q$ be emulated with limbs of width b bits inside a native field $\mathbb{F}_p$. Consider evaluating a degree- d polynomial via an inner product over the cached power vector $\mathcal{X}$. Since limbs are accumulated independently, each result-limb is a sum of $d + 1$ products of two b -bit values, so its magnitude is bounded by 2^β with $\beta = 2b + \lceil \log_2(d + 1) \rceil$.*

The deferred single-element reduction is safe whenever $\beta < \log_2 p$.

Corollary 1 (BN254 pairing instantiation). *For BN254 pairing verification: $b = 64$, $d \leq 256$, and $p \approx 2^{254}$. Then*

$$\beta = 2 \cdot 64 + \lceil \log_2 257 \rceil = 128 + 9 = 137 < 254,$$

so the deferred reduction is safe.

Beyond this static bound, the implementation tracks each element’s overflow at compile time and `InnerProductNoReduce` inserts a reduction whenever an accumulation would exceed the native capacity, so Lemma ?? is also enforced dynamically for rings and degrees other than the BN254 instantiation above.

7 Conclusion

We have shown that choosing the right granularity for polynomial identity testing—entire Miller loop steps rather than individual field multiplications—yields substantial and measurable improvements in arithmetized pairing verification. The key technique, collapsing a composite multi-operand product into a single Euclidean division check with batched quotient accumulation, is simple to implement and broadly applicable to other multi-step cryptographic computations in arithmetized environments.

Table 6: Performance improvements in Garaga: Cairo implementation on Starknet

MULMOD		ADDMOD		COMMITTS		VM STEPS	
Before	After	Before	After	Before	After	Before	After
BN254: 3-Pairing with Miller loop and final exponentiation							
19,142	12,656	21,686	15,142	5,689	2,807	338,678	212,902
33.9%		30.2%		50.7%		37.1%	
BLS12-381: 3-Pairing with Miller loop and final exponentiation							
16,484	11,287	20,669	15,420	5,421	3,167	325,757	219,155
31.5%		25.4%		41.6%		32.7%	
MULMOD/ADDMOD: Field operations in $\mathbb{F}_p$ via built-in function calls							
COMMITTS: Elements committed for Fiat-Shamir, see § ??							
VM STEPS: CairoVM Steps [?] (CPU Cycles)							

7.1 Applications and Impact

Our approach delivers maximum impact in resource-constrained execution environments where custom auxiliary data can be provided for cheaper verification. **Recursive proof systems** benefit the most, greatly reducing outer circuit complexity through fewer commitments and operations. **ZKVMs** (RISC Zero, SP1, Jolt) benefit naturally as they express computation as constraint systems compatible with our polynomial framework. **Calldata-constrained blockchains** (Ethereum L2s, BitVM) also benefit, as fewer commitments mean smaller proof transcripts and associated calldata costs.

This is already implemented in-production in Garaga [?] for CairoVM and is currently the standard for all pairing based ZK proofs on Starknet. The polynomial ring toolkit (Section ??) in gnark, submitted as a part of this work, will directly impact the `evmprecompiles/ecpair` precompile via `AssertMillerLoopAndFinalExpIsOne`.

7.2 Implementation and Performance

The Garaga implementation [?] in Cairo/Starknet (2024) sufficiently reduced the commitment overhead and modulo operations (Table ??) to fit Groth16 proof verification within Starknet’s transaction limits for the first time. However, Cairo [?] is a specialized environment with its own cost metrics, making direct comparison with mainstream ZK frameworks difficult. This work situates the same technique within gnark—the standard framework for field emulation in arithmetized circuits—providing a systematic benchmark against an industry baseline.

In Table ??, the *Before* column is Garaga verifying each extension field multiplication individually via the polynomial identity testing of [?]; the *After* column is Garaga using our consolidated relations. MULMOD/ADDMOD count modular field operations, COMMITTS count Poseidon [?] input commitments for Fiat-Shamir, and VM STEPS count CairoVM CPU cycles—together a low-level estimate of computational complexity inside the virtual machine.

Our polynomial ring toolkit implementation in gnark is benchmarked in Table ??, measured via a 3-pairing product check (see below) on BN254 and BLS12-381. We compare our results against a **gnark baseline** using the upstream `emulated` package [?], which represents the state-of-the-art for emulated pairings and includes the optimizations of [?] and [?].

The **This Work** numbers were obtained by compiling the exact same benchmark against our fork, replacing the baseline’s extension field multiplications with our new toolkit. No changes were made to the circuit definition or test harness. Both implementations

exploit the sparseness of the Miller line functions (Section ??). The excess memory savings come from the commitment and deferred-check pipeline: fewer deferred checks reduce both committed values and hint data payloads.

Benchmark circuit. Both implementations are evaluated against the same gnark circuit: a 3-pairing multi-pairing check $\prod_{i=1}^3 e(P_i, Q_i) = 1$ via `PairingCheck`. This corresponds to the pairing workload of a Groth16 verification, which reduces to a product-of-pairings check over three variable pairs once the constant $e(\alpha, \beta)$ term from the trusted setup is folded in. The same 3-pairing configuration is used in the CairoVM benchmarks of Table ??, allowing a like-for-like comparison across environments. Test circuit can be found at `sw_bn254` package in `g16_simulation_test.go` file in our implementation repo - <https://anonymous.4open.science/r/tiny-gnark-DDCD/>.

Test circuit can be found at `sw_bn254` package in `g16_simulation_test.go` file in our implementation repo - <https://anonymous.4open.science/r/tiny-gnark-DDCD/>.

7.3 Further Work

Several natural extensions of this work remain open. First, the framework can be readily extended to additional curves beyond BN254 and BLS12-381, notably **BW6-761** for recursive proof chain verification on Ethereum. A further direction is exploring the technique for native field pairings, such as BLS12-377 inside BW6-761 [?], where the trade-off between batching savings and commitment costs warrants dedicated analysis.

Finally, the underlying technique generalizes beyond pairings to any setting involving repeated operations over the same polynomial rings. Another interesting instance is lattice-based cryptography in ZK-SNARKs: schemes such as ML-KEM and ML-DSA operate in cyclotomic rings over $\mathbb{Z}_q[x]/(x^n + 1)$. Though the small moduli typical of lattice based schemes don't require active reductions, the polynomial identity testing would help in two significant ways. One, skip the need to do the in-circuit NTT transforms, and two, provide reduced coefficients ready for efficient packing for public inputs.

Exploring the cost-benefits of this approach in lattice-based ZK-SNARKs is another promising direction for future work.

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Table 7: Performance comparison: gnark implementation, 3-pairing product check on BN254 and BLS12–381

Metric	gnark Baseline	This Work	Red
BN254 – SCS (Plonkish) Constraint System			
Nb Constraints	1,592,440	1,035,550	
«««< HEAD «««< HEAD Nb $\mathbb{F}_q$ Committed	571,125	269,616	
===== Nb Commitments	571,125	269,616	
»»»> e1e4e0a (chore: update bench) ===== Nb $\mathbb{F}_q$ Committed	571,125	269,616	
»»»> fcab282 (fix: commitment terminology) Compile Time	615.04 ms	347.13 ms	
Compile Memory	1.33 GB	0.82 GB	
Solve Time	167.57 ms	97.30 ms	
Solve Memory	348.73 MB	181.24 MB	
BN254 – R1CS Constraint System			
Nb Constraints	491,447	316,167	
«««< HEAD «««< HEAD Nb $\mathbb{F}_q$ Committed	571,125	269,616	
===== Nb Commitments	571,125	269,616	
»»»> e1e4e0a (chore: update bench) ===== Nb $\mathbb{F}_q$ Committed	571,125	269,616	
»»»> fcab282 (fix: commitment terminology) Compile Time	617.63 ms	367.25 ms	
Compile Memory	1.60 GB	0.98 GB	
Solve Time	175.42 ms	106.36 ms	
Solve Memory	173.26 MB	112.72 MB	
BLS12–381 – SCS (Plonkish) Constraint System			
Nb Constraints	1,784,247	1,191,105	
«««< HEAD «««< HEAD Nb $\mathbb{F}_q$ Committed	660,160	300,104	
===== Nb Commitments	660,160	300,104	
»»»> e1e4e0a (chore: update bench) ===== Nb $\mathbb{F}_q$ Committed	660,160	300,104	
»»»> fcab282 (fix: commitment terminology) Compile Time	650.02 ms	410.48 ms	
Compile Memory	1.48 GB	0.97 GB	
Solve Time	180.05 ms	111.60 ms	
Solve Memory	362.14 MB	284.30 MB	
BLS12–381 – R1CS Constraint System			
Nb Constraints	545,510	368,434	
«««< HEAD «««< HEAD Nb $\mathbb{F}_q$ Committed	660,160	300,104	
===== Nb Commitments	660,160	300,104	
»»»> e1e4e0a (chore: update bench) ===== Nb $\mathbb{F}_q$ Committed	660,160	300,104	
»»»> fcab282 (fix: commitment terminology) Compile Time	686.95 ms	424.01 ms	
Compile Memory	1.74 GB	1.13 GB	
Solve Time	182.72 ms	121.15 ms	
Solve Memory	231.77 MB	118.61 MB	
«««< HEAD	Hardware: Apple M3 Pro (goarch:arm64), 36gb, macOS Tahoe 26.5.2		
$\mathbb{F}_q$ Committed: Elements committed via <code>api.Commit</code> calls for Fiat-Shamir, see § ??.			
=====	Hardware: Apple M3 Pro (goarch:arm64)		
$\mathbb{F}_q$ Committed: Elements committed via <code>api.Commit</code> calls for Fiat-Shamir.			
»»»> fcab282 (fix: commitment terminology) height			